AI DAILY LEARNING · DAY 111 · JULY 17, 2026

Meet the Model Router

Gemini 3.5 Pro Landed Today and Pushed the Live Model Count to Seven Price Tiers — Here's the Software That Now Picks Which One Answers Each Request

Google shipped Gemini 3.5 Pro today, July 17 — a full architectural rebuild with a 2-million-token context window and a Deep Think reasoning mode gated behind the $250-a-month Ultra tier. That's the seventh live price-and-capability tier across the frontier, joining GLM-5.2, DeepSeek V4, Qwen 3.6, Claude Sonnet 5, Grok 4.5, and GPT-5.6 — and picking the right one by hand, prompt by prompt, has quietly stopped being a job a person can do well.

Day 110 closed with a promise: the natural next move isn't picking a model, it's picking a router. Day 108 tracked a safety index where no lab scored above a C+; Day 109 watched three labs reprice within ten days; Day 110 found three open-weight releases that each win a different job. Today the axis flips again — from which model is best to which layer of software decides that for you, on every single call.

7	85%	14%	0-10
Live price-and-capability tiers across frontier and open-weight labs as of today's Gemini launch	RouteLLM's peer-reviewed cost cut on MT Bench, keeping 95% of GPT-4 Turbo-level quality	Share of queries RouteLLM's classifier actually sends to the strong, expensive model	OpenRouter's Auto Router dial — 0 always picks the best model, 10 always picks the cheapest

1) Seven tiers is one too many to pick by hand

Gemini 3.5 Pro arrives after Google reportedly scrapped an earlier version outright, after engineers found structural failures in recursive tool-calling and SVG generation, and rebuilt it on an entirely new pretraining run. The result ships a 2-million-token context window — double Gemini 3.5 Flash's 1 million — a Deep Think reasoning layer for multi-step problems, and autonomous workflow capability, at a reported $12-15 per million input tokens and $36-45 per million output tokens: roughly double Sonnet 5's introductory rate and in the same range as GPT-5.6 Terra.

That lands on top of a field that was already crowded. **GLM-5.2, DeepSeek V4,** and **Qwen 3.6** shipped inside the same ten-day window Day 110 covered; **Sonnet 5, Grok 4.5,** and **GPT-5.6** shipped the ten days before that. Seven models, seven different specialties, none of them dominant at everything — which means benchmarking every new release against your own workload before every project no longer scales. That's exactly the gap a routing layer exists to close.

2) How a routing layer actually decides

RouteLLM, built by the LMSYS team behind Chatbot Arena, trains lightweight classifiers — a matrix-factorization model and a BERT-based variant — on human preference data to predict which model wins a given prompt, then routes accordingly. It's open-source, peer-reviewed at ICLR 2025, and self-hosted, which means the routing logic lives in your own infrastructure rather than a vendor's black box.

OpenRouter's Auto Router, powered by NotDiamond, takes the hosted route: it analyzes each incoming prompt, selects a model from a curated pool, forwards the request, and returns metadata showing which model actually answered. It pins the chosen model and provider for the rest of a conversation to keep behavior consistent and prompt-cache hits intact, charges no fee of its own beyond the selected model's standard rate, and exposes a single cost_quality_tradeoff dial from 0 (always the most capable model) to 10 (always the cheapest). **Martian** takes a third approach — a proxy layer that routes in real time behind one model-agnostic endpoint, so the application never has to know which model actually ran.

The frontier is already moving past static classifiers: bandit-feedback online-learning routers like BaRP and PILOT adapt from live traffic instead of a training set frozen at launch, MCP Gateways are emerging as a single control plane in front of an entire tool-plus-model stack, and vLLM's Iris project is pushing the routing decision below the application layer entirely, into the inference engine itself.

3) The catch: routing is a new dependency, not a free lunch

RouteLLM's 85% figure is real but pair-specific — GPT-4 Turbo versus Mixtral 8x7B, scored on MT Bench. Run the same idea with a simpler BERT classifier on MMLU instead, and the savings drop to 45%. Martian's advertised "up to 98%" is explicitly an upper bound, not a typical result. None of these numbers travel to your workload unmodified; they're proof the technique works, not a guarantee of what you'll get.

A router that misjudges a prompt doesn't throw an error — it silently sends a hard question to a cheap model and returns a worse answer with total confidence. That makes the router's own accuracy something you now have to monitor, on top of monitoring the models it's choosing between. It also adds a dependency to the request path: if a hosted router's curated model pool or classifier goes stale, quality can drift without a single line of your own code changing. In production, teams that actually tune a routing layer report bill cuts in the 40-85% range — a wide band, because it depends entirely on how much of your traffic ever needed the expensive model in the first place.

THE ONE RULE

A router automates the judgment you'd apply by hand — it doesn't replace it. Benchmark the router's accuracy on a sample of your own traffic before you trust its savings number, the same way you'd benchmark any model before shipping it.

Tool	Type	Cost / savings	Best at
RouteLLM (LMSYS)	Open-source router	85% cost cut (MT Bench, 95% GPT-4 Turbo quality)	Self-hosted, research-grade routing
OpenRouter Auto Router	Hosted router (NotDiamond)	$0 routing fee -- pays selected model's rate	One-dial cost/quality tuning (0-10)
Martian	Hosted proxy router	Up to 98% claimed (upper bound)	Real-time routing with fallbacks
Gemini 3.5 Pro (ref., new today)	Frontier model	$12-15 in / $36-45 out per 1M	2M context, Deep Think reasoning

VIRAL / OPEN-SOURCE SPOTLIGHT

RouteLLM: the open-source router that turned model-picking into a solved problem

RouteLLM's GitHub repo is getting a fresh wave of attention this week, as teams staring down seven live price tiers go looking for something better than hand-picking a model per project. Its pitch is almost aggressively simple: train a lightweight classifier on human preference data, then let it decide, for each incoming prompt, whether a cheap model will do or a strong one is worth the money. The ICLR 2025 result behind it holds up — 85% lower cost on MT Bench while keeping 95% of GPT-4 Turbo-level quality, sending only 14% of traffic to the expensive model. It's popular for the same reason **OpenRouter's** Auto Router is spreading through developer circles this week: neither one asks a team to become a routing-research lab first. One is open-source and self-hosted, the other is a hosted dial — but both let you stop reading benchmark screenshots and start routing.

85%	45%	40-85%
RouteLLM's cost cut on MT Bench, keeping 95% of GPT-4 Turbo quality (ICLR 2025)	A simpler BERT classifier's cost cut on MMLU using the same idea -- the more conservative real-world number	The bill-reduction range teams report in production once a routing layer is actually tuned

MARKET SIGNAL

Today is also the opening day of Shanghai's World AI Conference, and for the first time since it launched in 2018, **Xi Jinping** is delivering the keynote in person — China's clearest signal yet that it wants to help write AI's governance rules, not just follow them, including a push for a Shanghai-headquartered World AI Cooperation Organization as a counterweight to Western-led governance bodies. It lands the same week Anthropic confirmed early talks with Samsung over custom Claude inference chips, aimed squarely at its own $1.25-billion-a-month compute bill. Put next to today's topic, it's the same pattern routing exists to manage: every lab, East or West, is racing to control the layer beneath the model — the chips, the governance rules, the routing decisions — because that's where the actual cost and control now live, not on the leaderboard.

PRACTICAL TAKEAWAYS

1. Don't trust a routing benchmark you haven't run on your own traffic

RouteLLM's 85% figure is real but specific to one model pair on one benchmark; a BERT-based version of the same idea only cleared 45% on a different one. Test any router against a sample of your actual prompts before trusting its savings number.

2. Start with a dial, not a black box

Tools like OpenRouter's Auto Router expose a single cost_quality_tradeoff parameter (0-10) instead of hiding the decision entirely. Keep that kind of manual override while you build confidence in a router's judgment, rather than adopting one you can't audit.

3. Budget for an eighth tier, then a ninth

Gemini 3.5 Pro is this week's new price tier; six labs shipped inside the last two weeks alone. Whatever routing setup you build should assume the model list keeps growing, not that today's seven options are the final list.

TOMORROW

Tomorrow — Day 112: Shanghai's World AI Conference runs through July 20, and Xi Jinping's keynote is only the opening move — China wants a Shanghai-headquartered World AI Cooperation Organization to sit alongside, or instead of, the Western-led governance bodies. We look at what WAICO actually proposes, why more than 140 forums and 1,100 exhibitors are betting on it, and what it would mean for a routing layer that, as of today, is already mixing U.S. and Chinese models in the same request path.
